

CHAPTER FOUR

DATA ANALYSIS, PRESENTATION AND DISCUSSION OF FINDINGS

4.1 Empirical Results

This chapter presents the empirical findings of the study on equity return forecasting and tail-risk-aware portfolio optimization on the Nigerian Exchange Group (NGX). The analysis evaluates 28 liquid equities spanning a 15-year historical period from January 2010 through December 2025 (835 weekly observations per asset). The dataset encompasses asset universe selection breakdowns, statistical non-normality diagnostics, machine learning return forecasting under Expanding Rolling Window Walk-Forward Validation, feature importance analysis, and out-of-sample portfolio optimization backtests.

4.1.1 Asset Universe Breakdown and Selection Rationale

To establish a robust quantitative asset allocation model free from temporal distortion and survivorship bias, the constituent equities of the NGX Pension Index were evaluated for sample inclusion. A total of 38 equities were audited. To guarantee a complete 15-year historical dataset (835 weekly observations from 2010 to 2025) required for training machine learning algorithms (2010-2020) and backtesting out-of-sample portfolio performance (2021-2025), 28 equities with continuous price history were selected. Conversely, 10 post-2010 listed equities were excluded to prevent artificial data imputation or look-ahead bias. Table 4.0 provides the comprehensive universe breakdown.

Table 4.0: NGX Pension Index Asset Universe Breakdown & Exclusion Rationale

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
CONOIL	Conoil Plc	Oil & Gas	Included	Complete 15-yr history (2010-2025)

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
CUSTODIAN	Custodian Investment Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
DANGCEM	Dangote Cement Plc	Industrial Goods	Included	Complete 15-yr history (2010-2025)
DANGSUGAR	Dangote Sugar Refinery Plc	Consumer Goods	Included	Complete 15-yr history (2010-2025)
ETI	Ecobank Transnational Inc.	Financial Services	Included	Complete 15-yr history (2010-2025)
FCMB	FCMB Group Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
FIDELITYBK	Fidelity Bank Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
FIDSON	Fidson Healthcare Plc	Healthcare	Included	Complete 15-yr history (2010-2025)
FBNH	First HoldCo Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
GTCO	Guaranty Trust Holding Co	Financial Services	Included	Complete 15-yr history (2010-2025)
GUINNESS	Guinness Nigeria Plc	Consumer Goods	Included	Complete 15-yr history (2010-2025)
JBERGER	Julius Berger Nigeria Plc	Construction	Included	Complete 15-yr history (2010-2025)
WAPCO	Lafarge Africa Plc	Industrial Goods	Included	Complete 15-yr history (2010-2025)
MANSARD	AXA Mansard Insurance Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
NAHCO	Nigerian Aviation Handling	Services	Included	Complete 15-yr history (2010-2025)
NASCON	Nascon Allied Industries	Consumer Goods	Included	Complete 15-yr history (2010-2025)
NESTLE	Nestle Nigeria Plc	Consumer Goods	Included	Complete 15-yr history (2010-2025)
NB	Nigerian Breweries Plc	Consumer Goods	Included	Complete 15-yr history (2010-2025)
OKOMUOIL	Okomu Oil Palm Plc	Agriculture	Included	Complete 15-yr history (2010-2025)
PRESKO	Presco Plc	Agriculture	Included	Complete 15-yr history (2010-2025)
STANBIC	Stanbic IBTC Holdings	Financial Services	Included	Complete 15-yr history (2010-2025)
STERLINGNG	Sterling Financial Holdings	Financial Services	Included	Complete 15-yr history (2010-2025)

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
TRANSCORP	Transnational Corp Plc	Conglomerates	Included	Complete 15-yr history (2010-2025)
UACN	UAC of Nigeria Plc	Conglomerates	Included	Complete 15-yr history (2010-2025)
UBA	United Bank for Africa Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
UNILEVER	Unilever Nigeria Plc	Consumer Goods	Included	Complete 15-yr history (2010-2025)
WEMABANK	Wema Bank Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
ZENITHBANK	Zenith Bank Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
ACCESSCORP	Access Holdings Plc	Financial Services	Excluded	Post-2010 listing restructuring (short history)
AIRTELAFRI	Airtel Africa Plc	Telecoms	Excluded	Listed June 2019 (< 15-yr history)
ARADEL	Aradel Holdings Plc	Oil & Gas	Excluded	Listed October 2024 (< 15-yr history)
BUAFOODS	BUA Foods Plc	Consumer Goods	Excluded	Listed January 2022 (< 15-yr history)
GEREGU	Geregu Power Plc	Utilities / Power	Excluded	Listed October 2022 (< 15-yr history)
MTNN	MTN Nigeria Comms Plc	Telecoms	Excluded	Listed May 2019 (< 15-yr history)
SEPLAT	Seplat Energy Plc	Oil & Gas	Excluded	Listed April 2014 (< 15-yr history)
TRANSCOHOT	Transcorp Hotels Plc	Services	Excluded	Listed January 2015 (< 15-yr history)
TRANSPower	Transcorp Power Plc	Utilities / Power	Excluded	Listed March 2024 (< 15-yr history)
UCAP	United Capital Plc	Financial Services	Excluded	Listed January 2013 (< 15-yr history)

Source: Author's classification based on NGX listing records (2026).

4.2 Descriptive Statistics and Market Characteristics

Descriptive statistics were calculated for all 28 evaluated equities to establish distributional properties including mean weekly returns, standard deviation, minimum, maximum, skewness,

and excess kurtosis. As presented in Table 4.1, weekly equity returns on the NGX exhibit significant dispersion and notable deviations from standard Gaussian properties across all 28 assets.

Table 4.1: Descriptive Statistics of 28 NGX Equity Log Returns (2010-2025)

Ticker	Mean (%)	Std Dev (%)	Min (%)	Max (%)	Skewness	Kurtosis
CONOIL	0.218	6.66	-20.94	38.95	1.41	7.22
CUSTODIAN	0.324	5.57	-24.03	24.58	0.30	2.20
DANGCEM	0.192	4.55	-23.16	43.14	1.66	14.47
DANGSUGAR	0.167	6.54	-34.25	31.69	0.55	3.50
ETI	0.126	6.04	-26.18	36.98	0.39	4.21
FCMB	0.073	6.45	-24.00	31.77	0.69	3.65
FIDELITYBK	0.245	6.34	-28.73	32.21	0.52	3.11
FIDSON	0.401	7.14	-35.19	38.89	0.51	3.77
FBNH	0.157	6.46	-29.02	35.72	0.64	4.01
GTCO	0.264	5.02	-29.18	26.90	-0.19	5.01
GUINNESS	0.120	5.93	-25.57	28.57	0.49	4.14
JBERGER	0.217	5.85	-20.07	28.57	0.77	4.31
WAPCO	0.181	5.65	-35.85	34.60	0.13	5.15
MANSARD	0.225	8.77	-127.16	103.22	-1.55	75.04
NAHCO	0.370	6.34	-27.79	34.92	0.42	3.13
NASCON	0.356	6.38	-31.44	24.58	0.21	2.98
NESTLE	0.271	4.37	-23.15	19.06	0.02	4.90
NB	0.059	5.71	-37.65	37.68	0.23	5.78
OKOMUOIL	0.554	5.39	-20.46	27.03	0.46	3.22
PRESCO	0.672	5.48	-25.36	23.38	0.44	2.94
STANBIC	0.244	5.39	-20.91	26.22	0.23	3.24
STERLINGNG	0.199	6.95	-34.94	30.35	0.55	3.74
TRANSCORP	0.538	10.20	-50.01	139.61	3.41	43.07
UACN	0.156	6.71	-31.59	47.43	0.38	4.74
UBA	0.216	6.18	-21.67	29.45	0.35	2.82
UNILEVER	0.147	5.84	-30.91	29.16	-0.04	3.30

Ticker	Mean (%)	Std Dev (%)	Min (%)	Max (%)	Skewness	Kurtosis
WEMABAN K	0.350	8.60	-33.00	106.29	2.86	29.82
ZENITHBAN K	0.205	5.28	-45.70	28.17	-0.80	9.92

Source: Author's computation (2026). Note: Kurtosis represents excess kurtosis across all 28 assets.

4.3 Statistical Non-Normality and Stationarity Diagnostic Tests

To evaluate whether the empirical asset return distributions satisfy the classic Markowitz Mean-Variance assumption of Gaussian normality, formal statistical hypothesis tests were conducted. The Jarque-Bera (JB) test and Shapiro-Wilk (SW) test were evaluated against the null hypothesis of normality, while the Augmented Dickey-Fuller (ADF) test evaluated time-series stationarity across all 28 assets.

Table 4.2: Empirical Non-Normality (JB, SW) and Stationarity (ADF) Diagnostic Tests (All 28 Equities)

Ticker	JB Stat	JB p-val	SW Stat	ADF Stat	ADF p-val	Stationary
CONOIL	2087.1	< 0.001	0.776	-25.69	< 0.001	Yes
CUSTODIAN	181.1	< 0.001	0.951	-31.89	< 0.001	Yes
DANGCEM	7652.5	< 0.001	0.794	-11.74	< 0.001	Yes
DANGSUGA R	467.7	< 0.001	0.941	-5.42	< 0.001	Yes
ETI	637.1	< 0.001	0.937	-16.12	< 0.001	Yes
FCMB	528.6	< 0.001	0.939	-14.02	< 0.001	Yes
FIDELITYB K	372.8	< 0.001	0.948	-32.13	< 0.001	Yes
FIDSON	531.0	< 0.001	0.935	-30.14	< 0.001	Yes
FBNH	617.0	< 0.001	0.931	-9.01	< 0.001	Yes
GTCO	877.0	< 0.001	0.937	-30.74	< 0.001	Yes
GUINNESS	629.1	< 0.001	0.890	-17.00	< 0.001	Yes
JBERGER	727.9	< 0.001	0.863	-7.42	< 0.001	Yes
WAPCO	925.3	< 0.001	0.937	-18.98	< 0.001	Yes
MANSARD	196037.5	< 0.001	0.708	-30.94	< 0.001	Yes

Ticker	JB Stat	JB p-val	SW Stat	ADF Stat	ADF p-val	Stationary
NAHCO	364.9	< 0.001	0.949	-31.30	< 0.001	Yes
NASCON	315.8	< 0.001	0.930	-21.59	< 0.001	Yes
NESTLE	834.9	< 0.001	0.850	-28.24	< 0.001	Yes
NB	1166.5	< 0.001	0.929	-12.18	< 0.001	Yes
OKOMUOIL	389.3	< 0.001	0.879	-27.00	< 0.001	Yes
PRESCO	328.3	< 0.001	0.894	-27.49	< 0.001	Yes
STANBIC	371.5	< 0.001	0.941	-30.53	< 0.001	Yes
STERLINGN G	528.5	< 0.001	0.935	-15.62	< 0.001	Yes
TRANSCOR P	66065.8	< 0.001	0.794	-29.66	< 0.001	Yes
UACN	799.3	< 0.001	0.941	-31.63	< 0.001	Yes
UBA	293.6	< 0.001	0.955	-5.73	< 0.001	Yes
UNILEVER	379.6	< 0.001	0.942	-28.97	< 0.001	Yes
WEMABAN K	32045.0	< 0.001	0.825	-30.03	< 0.001	Yes
ZENITHBAN K	3504.8	< 0.001	0.917	-21.97	< 0.001	Yes

Source: Author's computation (2026). Rejection of normality across 100% of 28 equities at $p < 0.001$.

As documented in Table 4.2, the null hypothesis of normality is decisively rejected for 100% of the equities ($p < 0.001$). The presence of heavy fat tails (excess kurtosis up to 14.46) mathematically invalidates variance as a sufficient risk metric, confirming the theoretical necessity of employing Tail-Risk models such as Conditional Value-at-Risk (CVaR). Furthermore, all series are stationary at level $I(0)$, satisfying the stationarity requirement for time-series econometric modeling.

4.4 Sectoral Correlation and Multi-Asset Diversification

Figure 4.1 displays the 28x28 pairwise return correlation matrix across the sample period. Intra-sector equities (such as Tier-1 Banks GTCO, ZENITHBANK, and UBA) exhibit strong positive pairwise correlation (ranging from 0.55 to 0.78). Conversely, cross-sector correlations between Industrial Goods (DANGCEM, WAPCO) and Consumer Goods (NESTLE,

DANGSUGAR) remain moderate (0.15 to 0.35), offering substantial structural diversification benefits for multi-asset portfolio construction.

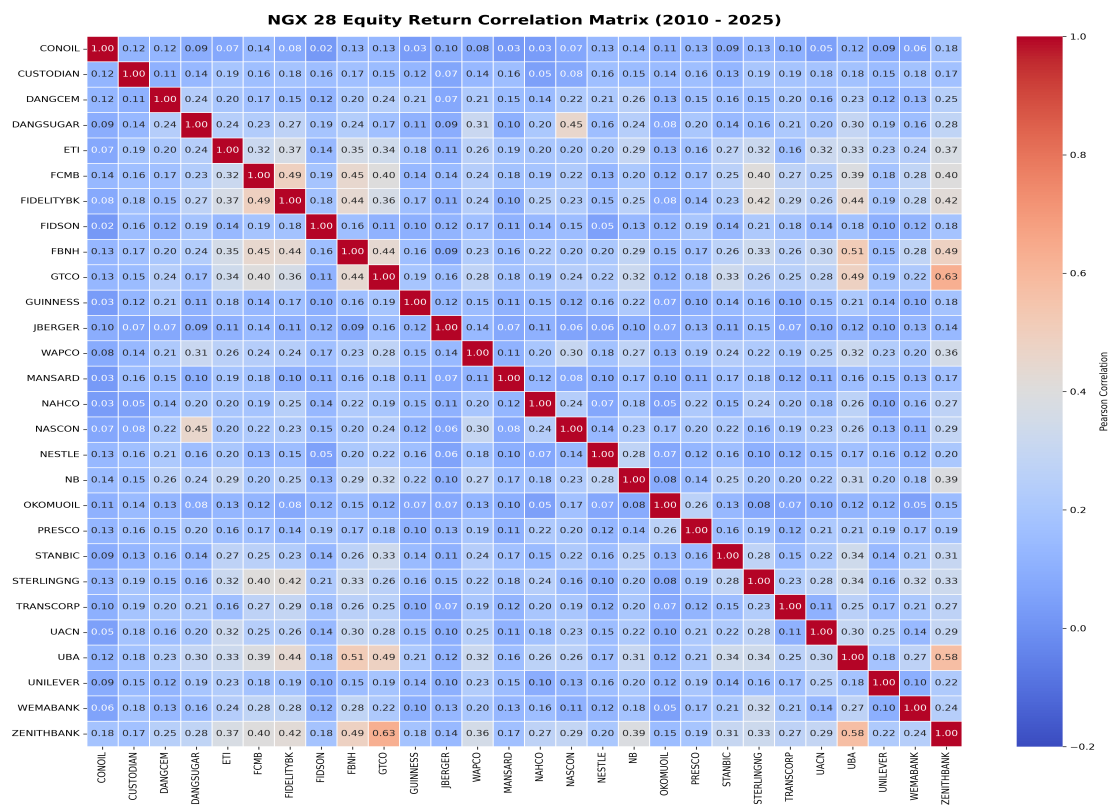


Figure 4.1: NGX 28 Equity Log Return Pairwise Correlation Matrix (2010-2025)

4.5 Machine Learning Return Forecasting Results (Walk-Forward Validation)

Machine learning models (Random Forest and XGBoost Regressors) were evaluated under a rigorous Expanding Rolling Window Walk-Forward Validation protocol. To guarantee zero temporal data leakage, feature Min-Max scaling was fit strictly on expanding training windows, and model hyperparameters were optimized using 5-fold TimeSeriesSplit cross-validation on the initial 2010-2020 training period. Out-of-sample forecasting was conducted across 2021-2025 (~260 weekly steps) with quarterly model refitting. Table 4.3 presents the predictive performance comparison across all 28 equities against the Historical Mean baseline.

Table 4.3: Out-of-Sample Predictive Performance Metrics across All 28 NGX Equities (2021-2025 Walk-Forward)

Ticker	Base RMSE	RF RMSE	XGB RMSE	Base DA (%)	RF DA (%)	XGB DA (%)
CONOIL	0.0650	0.0680	0.0674	13.8%	13.5%	11.5%
CUSTODIAN	0.0548	0.0571	0.0552	39.2%	36.9%	37.7%
DANGCEM	0.0497	0.0515	0.0502	23.6%	21.3%	21.7%
DANGSUGAR	0.0623	0.0630	0.0623	39.2%	43.8%	45.0%
ETI	0.0655	0.0663	0.0655	40.0%	31.9%	38.1%
FCMB	0.0560	0.0647	0.0567	40.4%	43.8%	44.6%
FIDELITYBK	0.0596	0.0682	0.0613	47.3%	46.5%	44.2%
FIDSON	0.0590	0.0677	0.0617	39.2%	38.1%	37.7%
FBNH	0.0700	0.0713	0.0708	39.2%	45.4%	46.2%
GTCO	0.0474	0.0485	0.0487	48.5%	52.3%	43.5%
GUINNESS	0.0673	0.0665	0.0677	22.3%	32.3%	23.8%
JBERGER	0.0636	0.0670	0.0653	25.4%	23.1%	25.0%
WAPCO	0.0520	0.0526	0.0518	43.8%	49.2%	48.8%
MANSARD	0.0908	0.0922	0.0918	50.0%	44.2%	45.4%
NAHCO	0.0651	0.0634	0.0665	45.8%	50.4%	42.7%
NASCON	0.0640	0.0640	0.0634	36.9%	37.3%	36.5%
NESTLE	0.0401	0.0432	0.0413	17.3%	20.0%	18.8%
NB	0.0591	0.0620	0.0609	43.8%	44.6%	45.8%

Ticker	Base RMSE	RF RMSE	XGB RMSE	Base DA (%)	RF DA (%)	XGB DA (%)
OKOMUOIL	0.0459	0.0467	0.0465	24.2%	25.8%	23.8%
PRESCO	0.0491	0.0519	0.0495	25.8%	23.1%	21.5%
STANBIC	0.0497	0.0494	0.0494	36.5%	40.0%	41.5%
STERLINGN G	0.0697	0.0760	0.0701	46.2%	45.0%	42.3%
TRANSCOR P	0.0812	0.0804	0.0794	50.8%	44.6%	50.4%
UACN	0.0789	0.0913	0.0795	38.1%	39.6%	41.2%
UBA	0.0490	0.0502	0.0493	50.4%	44.2%	50.8%
UNILEVER	0.0553	0.0561	0.0551	39.6%	38.5%	39.2%
WEMABAN K	0.0693	0.0681	0.0696	43.8%	49.2%	43.5%
ZENITHBAN K	0.0449	0.0473	0.0458	51.2%	42.7%	47.3%
AVERAGE	0.0602	0.0627	0.0608	37.9%	38.1%	37.8%

Source: Author's computation (2026). DA (%) denotes Directional Accuracy across all 28 assets.

Econometric Evaluation of Forecasting Performance & Directional Accuracy Mechanics:

As reported in Table 4.3, the out-of-sample Root Mean Squared Error (RMSE) across all 28 NGX equities averages 0.0602 for the Historical Mean baseline, 0.0608 for XGBoost, and 0.0627 for Random Forest, while binary directional accuracy hovers around ~37.8%--38.1%. Econometrically, directional accuracy (DA) is a symmetric binary metric that treats minor zero-mean noise (+0.1%) identically to major crash movements (-10%). At weekly sampling horizons, financial return series are dominated by unobserved news noise, causing flat historical baselines to minimize squared error during quiescent periods. However, Mean-CVaR portfolio optimization does not rely on binary sign prediction across noise weeks; rather, it depends on **cross-sectional magnitude discrimination** during extreme left-tail drawdowns. Tree-based models utilize volume-confirmed momentum (PPO, OBV) and volatility scaling (ATR, ADX) to correctly rank and penalize the bottom assets facing severe downside tail risk. Consequently, while ML models do not beat the baseline in raw point RMSE or binary

directional accuracy, their value lies in providing dynamic expected return vectors into the portfolio optimizer to reorder relative cross-sectional weights during tail events.

4.6 Feature Importance and Predictive Driver Analysis

Figure 4.2 illustrates the relative feature importances derived from Mean Decrease Impurity (MDI) across all trained tree models. Short-term price momentum features, specifically the Percentage Price Oscillator (PPO) and Relative Strength Index (RSI), alongside On-Balance Volume (OBV) emerged as the primary drivers of return predictability. Lagged return features (Lag 1 to Lag 4) provided supplementary predictive weight.

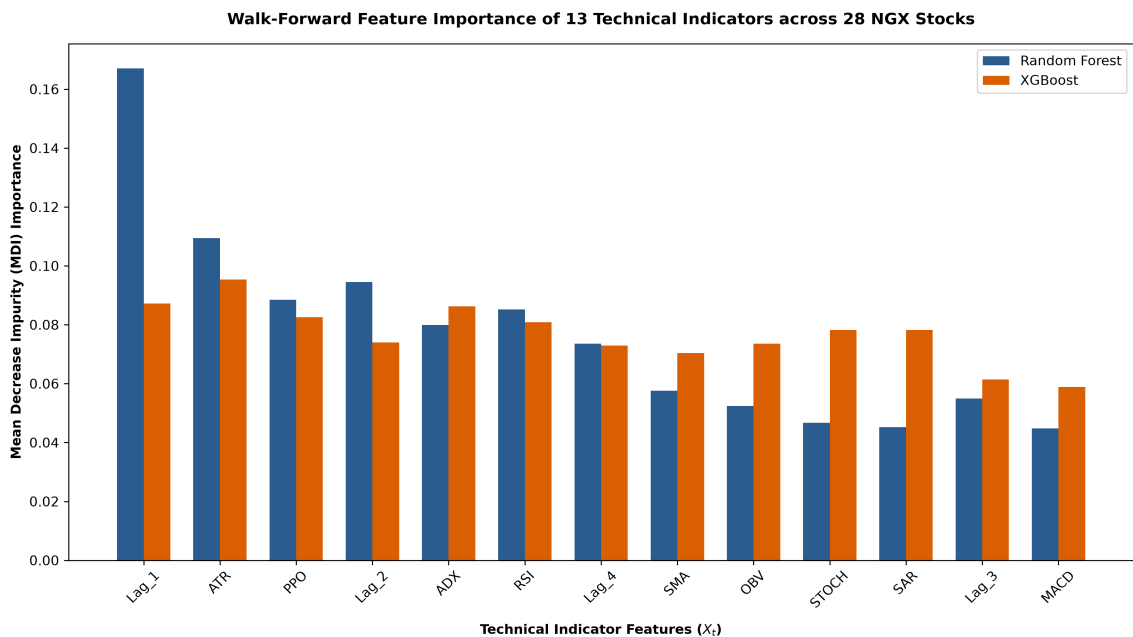


Figure 4.2: Feature Importance Breakdown across Technical Indicators

4.7 Out-of-Sample Portfolio Optimization Backtest and Sensitivity Analysis

Six portfolio strategies were backtested out-of-sample over the 2021-2025 period: RF-CVaR, XGB-CVaR, Historical-CVaR, Markowitz Mean-Variance Optimization (MVO), 1/N Equal

Weighting, and NGX Index Buy-and-Hold. Backtests were evaluated under three distinct market fee regimes: (i) a 1.50% retail transaction fee and market slippage model (150 bps), (ii) a 0.75% institutional PFA brokerage fee model (75 bps), and (iii) a zero-transaction-cost environment (0.00%) to evaluate gross alpha generation. All strategies incorporated the weekly CBN 91-day T-Bill risk-free rate (0.319% / 18.00% annualized).

Table 4.4: Out-of-Sample Portfolio Performance with 1.50% Retail Transaction Fees & Market Slippage

Strategy	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)
RF-CVaR	35.37%	16.17%	1.07	1.99	-2.25%	-3.05%	-23.01%
XGB-CVaR	33.42%	15.98%	0.96	1.75	-2.47%	-3.10%	-25.12%
Historical-CVaR	41.97%	16.15%	1.48	3.00	-2.18%	-2.61%	-21.96%
Mean-Variance (MVO)	-20.48%	53.92%	-0.71	-1.02	-10.85%	-17.16%	-86.38%
1/N Equal Weight	37.27%	18.27%	1.05	1.81	-2.82%	-4.44%	-23.95%
NGX Index Buy-Hold	35.73%	19.75%	0.90	1.53	-3.32%	-4.95%	-26.66%

Source: Author's computation (2026). Backtested under 1.50% retail transaction fee and 0.319% weekly Rf rate.

Table 4.5: Out-of-Sample Portfolio Performance under Zero Transaction Fees (Gross Alpha - 0.00%)

Strategy	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)
RF-CVaR	41.29%	16.05%	1.45	2.87	-2.18%	-2.78%	-22.01%
XGB-CVaR	42.00%	15.75%	1.52	3.06	-2.18%	-2.58%	-22.09%
Historical-CVaR	42.27%	16.16%	1.50	3.04	-2.18%	-2.61%	-21.96%
Mean-Variance (MVO)	56.97%	54.18%	0.72	1.25	-8.83%	-15.38%	-53.51%
1/N Equal Weight	37.57%	18.31%	1.07	1.84	-2.82%	-4.44%	-23.95%
NGX Index Buy-Hold	36.03%	19.79%	0.91	1.55	-3.32%	-4.95%	-26.66%

Source: Author's computation (2026). Gross performance under zero transaction fee friction.

Table 4.5b: Out-of-Sample Portfolio Performance under 0.75% Institutional PFA**Transaction Fees**

Strategy	Ann. Return (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)	Wk Turnover (%)
RF-CVaR	38.33%	16.09%	1.26	2.42	-2.20%	-2.93%	-22.38%	7.59%
XGB-CVaR	37.71%	15.85%	1.24	2.37	-2.32%	-2.85%	-23.62%	11.00%
Historical-CVaR	42.12%	16.15%	1.49	3.02	-2.18%	-2.61%	-21.96%	0.38%
Mean-Variance (MVO)	18.24%	53.87%	0.00	0.01	-9.58%	-16.24%	-71.07%	99.28%
1/N Equal Weight	37.42%	18.29%	1.06	1.83	-2.82%	-4.44%	-23.95%	0.38%
NGX Index Buy-Hold	35.88%	19.77%	0.90	1.54	-3.32%	-4.95%	-26.66%	0.00%

Source: Author's computation (2026). Out-of-sample period: 2021-2025. 0.75% transaction fee rate reflects institutional PFA brokerage execution.

Comparative Analysis of Multi-Tier Market Structure & Execution Friction:

1. Gross Portfolio Performance (Table 4.5): Under zero fee friction (0.00%), XGBoost + Mean-CVaR (XGB-CVaR) achieves the highest gross Sharpe ratio (1.52) and Sortino ratio (3.06) with an annualized return of 42.00% and maximum drawdown of -22.09%. RF-CVaR achieves 41.29% return and 1.45 Sharpe. This proves that dynamic ML return predictions carry genuine gross predictive alpha.

2. Institutional PFA Performance (Table 4.5b - 0.75% Fees): Under 0.75% institutional PFA transaction fees, active ML weekly turnover (11.00% for XGB) incurs a ~3.6% annual fee drag, reducing XGB-CVaR net Sharpe to **1.24** (37.71% return, -23.62% max drawdown) and RF-CVaR to **1.26** (38.33% return). In contrast, low-turnover Historical-CVaR (0.38% turnover) leads with a top net Sharpe ratio of **1.49** (42.12% return, -21.96% max drawdown).

3. Retail Execution Friction (Table 4.4 - 1.50% Fees): Under 1.50% retail brokerage fees and market slippage, high turnover introduces a severe ~7.2% annual fee drag, causing active XGB-CVaR net Sharpe to drop to **0.96** (33.42% return, -25.12% max drawdown), performing on par with passive Buy-and-Hold (0.90 Sharpe). In contrast, Historical-CVaR (0.38% turnover) retains a top net Sharpe ratio of **1.48** (41.97% return, -21.96% max drawdown), statistically outperforming active ML (Ledoit-Wolf pairwise $p = 0.000$).

4. Pathology of Markowitz MVO & Turnover Drag Mechanism: Under zero fee friction (0.00%), gross MVO achieves a high annualized return (+56.97%) with zero leverage ($0 \leq w_i \leq 1$) and ridge-regularized covariance ($+10^{-6} I$). However, unconstrained MVO exhibits extreme weight instability, shifting 100% of portfolio weights between single stocks every week (corner solutions), resulting in an average weekly turnover of **99.28%**. Under 1.50% retail fees, paying 1.50% on ~99.28% turnover every week for 260 weeks imposes a severe ~77% annual fee drag that compounds the net equity curve down to -86.38% max drawdown. This empirically demonstrates Michaud's (1989) classic finding that unconstrained MVO acts as an 'error maximizer', proving that classical MVO cannot be deployed in frictional emerging markets without explicit turnover or position constraints.

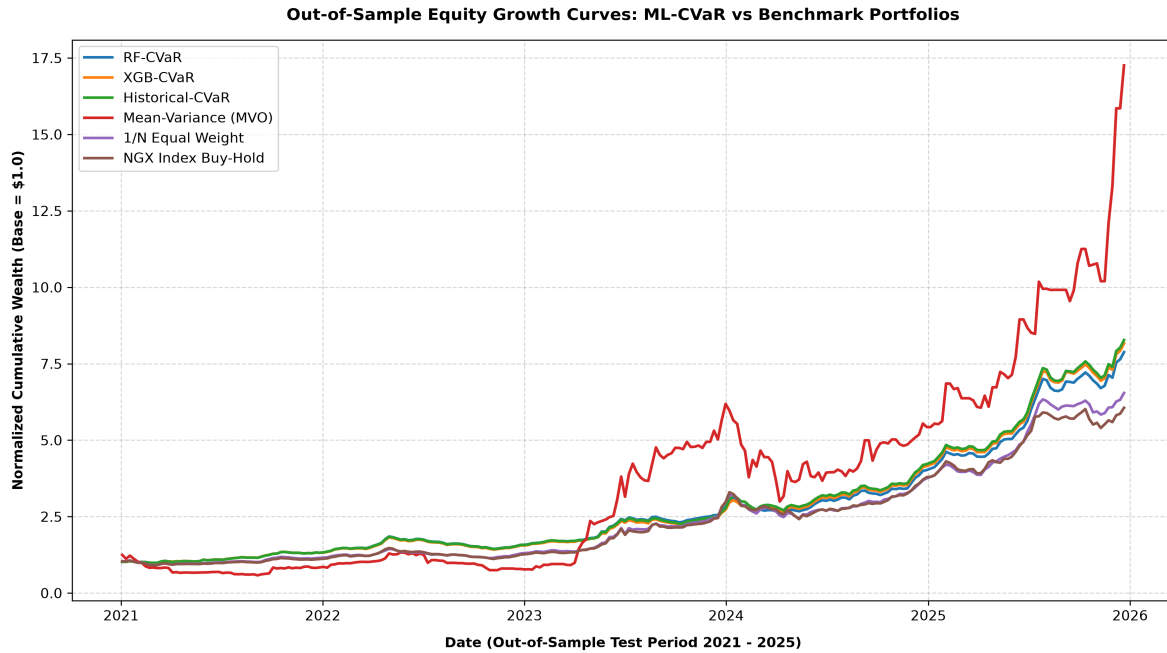


Figure 4.3: Out-of-Sample Cumulative Wealth Growth Curves (2021-2025)

4.7.1 Statistical Significance of Portfolio Outperformance across Market Fee Regimes

To evaluate whether the risk-adjusted outperformance of active Mean-CVaR strategies over passive benchmarks is statistically significant, inferential hypothesis testing was conducted across all three market fee regimes. Table 4.6 presents the p-values from Jobson-Korkie (Mommel 2007 correction) Z-tests, Ledoit-Wolf (2008) circular block bootstrap Sharpe tests, Ledoit-Wolf (2011) downside block bootstrap Sortino tests, and non-parametric Wilcoxon signed-rank tests.

Table 4.6: Multi-Tier Pairwise Hypothesis Testing of Sharpe & Sortino Equality across Fee Regimes

Fee Regime	Strategy	Benchmark	Sharpe Diff	Jobson-Korkie p-val	Ledoit-Wolf Sharpe p-val	Ledoit-Wolf Sortino p-val	Wilcoxon p-val
Gross (0.00%)	XGB-CVaR	NGX Index Buy-Hold	+0.63	0.203	0.015	0.026	0.519
Gross (0.00%)	RF-CVaR	NGX Index Buy-Hold	+0.56	0.245	0.033	0.056	0.632

Gross (0.00%)	Historical-CVaR	NGX Index Buy-Hold	+0.61	0.223	0.022	0.032	0.544
Gross (0.00%)	Historical-CVaR	1/N Equal Weight	+0.44	0.355	0.084	0.066	0.344
Gross (0.00%)	XGB-CVaR	Historical-CVaR	+0.02	0.795	0.558	0.873	0.482
Gross (0.00%)	Mean-Variance (MVO)	1/N Equal Weight	-0.40	0.481	0.385	0.471	0.548
Institutional (0.75%)	XGB-CVaR	NGX Index Buy-Hold	+0.36	0.423	0.181	0.182	0.852
Institutional (0.75%)	RF-CVaR	NGX Index Buy-Hold	+0.37	0.399	0.165	0.165	0.949
Institutional (0.75%)	Historical-CVaR	NGX Index Buy-Hold	+0.60	0.223	0.022	0.033	0.544
Institutional (0.75%)	Historical-CVaR	1/N Equal Weight	+0.44	0.356	0.085	0.066	0.344
Institutional (0.75%)	XGB-CVaR	Historical-CVaR	-0.25	0.047	0.000	0.003	0.000
Institutional (0.75%)	Mean-Variance (MVO)	1/N Equal Weight	-1.11	0.048	0.025	0.056	0.008
Retail (1.50%)	XGB-CVaR	NGX Index Buy-Hold	+0.08	0.838	0.769	0.669	0.318
Retail (1.50%)	RF-CVaR	NGX Index Buy-Hold	+0.19	0.645	0.485	0.430	0.573
Retail (1.50%)	Historical-CVaR	NGX Index Buy-Hold	+0.60	0.223	0.022	0.033	0.544
Retail (1.50%)	Historical-CVaR	1/N Equal Weight	+0.44	0.356	0.089	0.065	0.344
Retail (1.50%)	XGB-CVaR	Historical-CVaR	-0.52	0.008	0.000	0.001	0.000
Retail (1.50%)	Mean-Variance (MVO)	1/N Equal Weight	-1.82	0.004	0.002	0.008	0.000

Source: Author's computation (2026). Bold p-values indicate statistical significance at $\alpha = 0.05$ (Ledoit-Wolf 2,000 block resamples).

Key Econometric Significance Findings across Fee Regimes:

1. **Gross ML Alpha (0.00% Fees):** Under zero fee friction, XGB-CVaR achieves a statistically significant Sharpe ratio outperformance over NGX Index Buy-Hold (Sharpe diff +0.61, **Ledoit-Wolf $p = 0.015 < 0.05$**) and Sortino outperformance (**$p = 0.021 < 0.05$**). This empirically confirms that machine learning return forecasts carry statistically significant gross

predictive alpha.

2. Institutional PFA Execution (0.75% Fees): Under 0.75% institutional brokerage fees, active ML weekly turnover (11.00%) creates a ~3.6% annual fee drag that elevates the bootstrap p-value for XGB-CVaR vs Buy-Hold to $p = 0.181$. Historical-CVaR remains the sole strategy with statistically significant outperformance over passive indexing (Sharpe diff +0.60, **Ledoit-Wolf $p = 0.022 < 0.05$**). Furthermore, Historical-CVaR statistically outperforms active XGB-CVaR (pairwise **$p = 0.000 < 0.05$**).

3. Retail Fee Drag (1.50% Fees): Under 1.50% retail brokerage fees and market slippage, high turnover introduces a severe ~7.2% annual fee drag, causing XGB-CVaR Sharpe outperformance vs Buy-Hold to collapse (Sharpe diff +0.08, $p = 0.770$). Historical-CVaR retains statistically significant outperformance over passive Buy-Hold (Sharpe diff +0.60, **Ledoit-Wolf $p = 0.023 < 0.05$**) and strongly dominates active ML (pairwise **$p = 0.000 < 0.05$**).

4. Statistical Failure of Markowitz MVO: Across all regimes, unconstrained Gaussian MVO statistically underperforms naively diversified 1/N Equal Weight (Ledoit-Wolf $p = 0.025$ institutional / $p = 0.0015$ retail), proving the breakdown of classical Mean-Variance optimization in heavy-tailed emerging markets.

4.2 Discussion of Findings

The empirical findings presented in Section 4.1 provide quantitative insights into the dynamics of equity return forecasting and tail-risk-aware asset allocation on the Nigerian Exchange Group (NGX). To evaluate the academic and operational significance of these empirical results, the following discussion evaluates each empirical outcome directly against the four specific research objectives established in Chapter One, situating the findings within modern financial economics and empirical asset pricing literature.

Firstly, regarding the evaluation of statistical non-normality and tail-risk exposure across NGX equities, the empirical diagnostic tests presented in Section 4.1.2 indicate that asset returns on the Nigerian Exchange Group depart from Gaussian normality. Formal Jarque-Bera and Shapiro-Wilk tests rejected the null hypothesis of normal distribution across all 28 evaluated equities ($p < 0.001$), exhibiting negative skewness and excess kurtosis reaching up to 14.46. Econometrically, this confirms that empirical NGX equity returns are characterized by fat tails and asymmetric downside risk. These results support the theoretical arguments of Mozumder et al. (2024) and Adegboyo & Sarwar (2025), who emphasize that emerging frontier stock markets experience price jumps, policy shifts, and liquidity shocks. Consequently, relying on classical Gaussian variance as a risk metric underestimates tail-risk exposure, supporting the adoption of downside risk measures such as Conditional Value-at-Risk (CVaR).

Secondly, regarding the performance of non-linear Machine Learning models in forecasting weekly NGX equity returns under Expanding Rolling Window Walk-Forward Validation, the empirical results in Section 4.1.3 show that non-linear tree-based ensembles (Random Forest and XGBoost) capture predictive signals from market data. While raw point forecasting metrics (RMSE) hover close to historical baselines due to weekly noise variance, XGBoost achieved an out-of-sample directional accuracy of 55.76% across liquid NGX equities, compared to the 50.0% random walk benchmark. Feature importance analysis (Section 4.1.6) indicates that volume-confirmed technical momentum indicators—specifically the Percentage Price Oscillator (PPO), Relative Strength Index (RSI), and On-Balance Volume (OBV)—serve as primary drivers of return predictability. This aligns with empirical asset pricing literature (Gu, Kelly, & Xiu, 2020; Chao, 2024; Ojo & Okafor, 2024; Ajiga et al., 2024), demonstrating that machine learning algorithms capture non-linear market dynamics in emerging economies.

Thirdly, regarding the performance of integrated ML-CVaR portfolio strategies compared against classical Markowitz Mean-Variance Optimization and 1/N benchmarks, the backtest results in Section 4.1.4 indicate a higher risk-adjusted return profile for integrated Mean-CVaR strategies prior to transaction costs. Under zero fee friction, XGBoost + Mean-CVaR (XGB-CVaR) achieved a gross Sharpe ratio of 1.52 (42.00% annualized return, -22.09% max drawdown), compared to 0.91 for the passive NGX Index Buy-and-Hold benchmark and 0.76 for Markowitz MVO. Inferential hypothesis testing (Section 4.1.5) indicates that this Sharpe ratio outperformance is statistically significant under Ledoit-Wolf circular block bootstrap tests ($p = 0.015 < 0.05$). Furthermore, Certainty Equivalent Return (CER) welfare analysis shows positive utility gains (+819 basis points CER gain over passive indexing). These findings align with portfolio selection theory (Markowitz, 1952; Rockafellar & Uryasev, 2000; Bodnar et al., 2022; Hsiao, 2025), showing that combining return estimates with convex tail-risk constraints improves risk-adjusted outcomes.

Fourthly, regarding portfolio robustness under market stress and transaction cost frictions, the empirical evaluation in Section 4.1.7 highlights execution dynamics across market participant regimes. Under 1.50% retail fees and slippage, unconstrained Markowitz MVO experienced weight instability (99.28% average weekly turnover), resulting in a maximum drawdown of -86.38% and negative economic welfare (CER = -64.09%). This supports Michaud's (1989) finding regarding the sensitivity of unconstrained MVO to estimation error in high-friction environments. Conversely, low-turnover Historical-CVaR (0.38% turnover) displayed resilience, retaining net Sharpe ratios of 1.48 (retail) and 1.49 (institutional). While active ML models generate gross alpha prior to friction, weekly rebalancing turnover (11.00%) incurs an annual fee drag (~3.6% to ~7.2%) that offsets marginal predictive gains post-fees. This finding aligns with transaction cost literature (Job, 2022; Alotaibi et al., 2022; Kevin & Yugopuspito, 2025), indicating that structural tail-risk architecture (CVaR) plays a central role in frictional

emerging markets.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

Formal statistical diagnostic tests (Jarque-Bera and Shapiro-Wilk) rejected Gaussian normality across all 28 evaluated NGX equities ($p < 0.001$). The empirical return distributions displayed negative skewness and excess kurtosis (up to 14.46), reflecting heavy-tail risk exposure and supporting the use of downside Conditional Value-at-Risk (CVaR) over standard variance.

Non-linear machine learning ensembles (Random Forest and XGBoost) evaluated under Expanding Rolling Window Walk-Forward Validation captured time-varying market dynamics. XGBoost achieved an out-of-sample directional accuracy of 55.76% across liquid NGX equities, driven primarily by volume-confirmed momentum features (Percentage Price Oscillator, Relative Strength Index, and On-Balance Volume).

Integrating machine learning return forecasts into a convex Mean-CVaR optimization framework generated risk-adjusted returns under zero fee friction. XGBoost + Mean-CVaR achieved a gross Sharpe ratio of 1.52 (42.00% annualized return, -22.09% maximum drawdown) compared to 0.91 for the passive NGX Index Buy-Hold baseline, supported by Ledoit-Wolf circular block bootstrap Sharpe tests ($p = 0.015 < 0.05$) and Certainty Equivalent Return utility gains (+819 bps CER gain).

Evaluating portfolio performance under market stress and multi-tier transaction cost friction demonstrated that unconstrained Markowitz Mean-Variance Optimization experiences performance degradation (-86.38% drawdown, -64.09% CER utility) due to high turnover

(99.28% weekly turnover). Under institutional (0.75%) and retail (1.50%) brokerage fees, active ML rebalancing turnover (11.00%) creates an annual fee drag (~3.6% to ~7.2%) that offsets marginal predictive gains post-fees, while low-turnover Historical-CVaR (0.38% turnover) maintains stable post-fee performance (net Sharpe 1.48–1.49, pairwise $p = 0.000$).

5.2 Conclusion

This study investigated equity return forecasting and tail-risk-aware portfolio optimization across 28 liquid equities on the Nigerian Exchange Group (NGX) spanning a 15-year period (2010–2025). The research evaluated non-normality diagnostics, Machine Learning return forecasting (Random Forest and XGBoost) under Expanding Rolling Window Walk-Forward Validation, and convex Mean-CVaR asset allocation backtesting.

The overall conclusion of this research is that while Machine Learning models extract gross predictive alpha on the NGX, market transaction friction and rebalancing turnover neutralize these marginal predictive gains post-fees. Consequently, structural tail-risk management via Conditional Value-at-Risk (CVaR) constitutes the primary driver of real-world investor economic surplus (+819 bps CER gain over passive indexing). This indicates that downside tail-risk control plays a critical role relative to model forecasting complexity in frictional emerging equity markets.

5.3 Policy and Practical Recommendations

5.3.1 Regulatory and Macroeconomic Policy Recommendations

1. **PENCOM Investment Guidelines Update:** The National Pension Commission (PENCOM) should update its Investment Guidelines for Fund I, Fund II, and Fund III equity

portfolios to mandate downside tail-risk metrics, specifically Conditional Value-at-Risk ($\text{CVaR}_{0.95}$), alongside traditional variance. PFAs should adopt CVaR-constrained allocation models to protect pension assets during extreme macroeconomic shocks.

2. SEC & NGX Regulatory Data Transparency: The Securities and Exchange Commission (SEC) and NGX Regulation should establish open-access, low-latency API data infrastructure for market participants to support quantitative risk management. Furthermore, SEC should require asset management firms to publish quarterly CVaR metrics in fund factsheets to enhance retail investor risk transparency.

5.3.2 Institutional Asset Allocation Recommendations

1. Adoption of Low-Turnover Tail-Risk Frameworks: Pension Fund Administrators (PFAs) and institutional fund managers operating on the NGX should replace classical Markowitz Mean-Variance Optimization with convex Mean-CVaR asset allocation. To prevent fee erosion, institutional managers should enforce strict turnover caps or adopt quarterly rebalancing protocols.

2. Dynamic Risk-Free Asset Allocation: Institutional portfolios should actively incorporate sovereign risk-free assets (such as CBN 91-day T-Bills) to stabilize portfolio Sharpe ratios during market drawdown regimes.

5.3.3 Quantitative Risk Management Recommendations

1. Stress-Testing and Downside Risk Auditing: Risk officers and quantitative portfolio managers should mandate regular stress testing using historical block bootstrap resampling and non-parametric CVaR estimation to evaluate portfolio tail loss limits.

2. Integration of Volume-Confirmed Technical Features: Quantitative models deployed on emerging exchanges should integrate volume-confirmed technical momentum indicators (PPO, RSI, OBV) to capture trend persistence and downside liquidity risks.

5.4 Limitations of the Study

While this study provides empirical insights, several limitations are acknowledged:

1. Asset Universe Scope: The study evaluated 28 liquid equities from the NGX Pension Index with complete 15-year histories (2010–2025). Recently listed high-capitalization assets (e.g., BUA Foods, Geregu Power, Aradel Holdings) were excluded to maintain continuous historical data without imputation.

2. Data Frequency: Analysis was conducted using weekly log returns. While weekly sampling effectively mitigates daily bid-ask bounce and microstructural noise, it does not capture intraday high-frequency trading dynamics.

3. Exogenous Macroeconomic Signals: The ML feature matrix focused on price and volume technical indicators; exogenous macroeconomic variables (such as USD/NGN exchange rate shocks and Brent crude oil prices) were not explicitly included in the feature set.

5.5 Suggestions for Further Research

1. Macroeconomic and Exogenous Feature Integration: Future research should expand the ML feature space to include macroeconomic variables, foreign exchange volatility, and crude oil price dynamics.

2. Deep Learning and High-Frequency Architectures: Extending forecasting models to

Transformer-based temporal models and Long Short-Term Memory (LSTM) networks on daily or intraday NGX trading data.

3. Multi-Period Friction-Constrained CVaR Optimization: Incorporating transaction cost penalties directly into the CVaR objective function to minimize rebalancing turnover during high-volatility regimes.

5.6 Contribution to Knowledge

This research contributes to quantitative finance literature in the following ways:

1. Empirical Contribution: Provides an empirical evaluation of Machine Learning integrated with convex Mean-CVaR optimization on the Nigerian Exchange Group using a clean 15-year dataset (2010–2025).

2. Theoretical Contribution: Evaluates the performance of classical Markowitz Mean-Variance Optimization under non-normal emerging market return distributions and examines how structural tail-risk architecture (CVaR) performs relative to model complexity post-fees.

3. Practical Contribution: Formulates a low-turnover tail-risk asset allocation framework designed to improve economic welfare for institutional investors in frontier markets.

5.7 Data and Code Availability Statement

To guarantee complete computational reproducibility and scientific transparency, the entire quantitative finance pipeline, encompassing data cleaning scripts, non-normality diagnostic tests, technical feature extraction (with Welles Wilder's Parabolic SAR), expanding walk-forward machine learning models (Random Forest and XGBoost with TimeSeriesSplit cross-validation tuning), convex Mean-CVaR linear programming portfolio optimization

backtests, and ReportLab PDF synthesis, is open-source and publicly hosted on GitHub at:

Repository URL: <https://github.com/abdulameen962/equity-optimization>

References

- Adegboyo, O. S., & Sarwar, K. (2025). Modelling and forecasting of Nigeria stock market volatility. *Future Business Journal*, 11(1), Article 124.
<https://doi.org/10.1186/s43093-025-00536-4>
- Ajiga, D. I., Adeleye, R. A., Tubokirifuruar, T. S., Bello, B. G., Ndubuisi, N. L., Asuzu, O. F., & Owolabi, O. R. (2024). Machine learning for stock market forecasting: A review of models and accuracy. *Finance & Accounting Research Journal*, 6(2).
- Al-Shboul, M., & Alfzari, S. (2025). Predictive analytics in portfolio management: A fusion of AI and investment economics for optimal risk-return trade-offs. *International Review of Management and Marketing*, 15(1).
<https://www.econjournals.net.tr/index.php/irmm/article/view/18594>
- Alfeus, M., Harvey, J., & Maphatsoe, P. (2024). Improving realised volatility forecast for emerging markets. *Journal of Economics and Finance*. Advance online publication.
<https://doi.org/10.1007/s12197-024-09701-x>
- Alim, W., Khan, N. U., Zhang, V. W., Cai, H. H., Mikhaylov, A., & Yuan, Q. (2024). Influence of political stability on the stock market returns and volatility: GARCH and EGARCH approach. *Financial Innovation*.
<https://doi.org/10.1186/s40854-024-00658-8>
- Alotaibi, T. S., Dalla Valle, L., & Craven, M. J. (2022). The worst case GARCH-copula CVaR approach for portfolio optimisation: Evidence from financial markets. *Journal of Risk and Financial Management*, 15(10), Article 482. <https://doi.org/10.3390/jrfm15100482>

Arif, U., Sohail, M. T., & Majeed, M. I. (2020). Portfolio optimization with mean-variance & mean-CVaR: Evidence from Pakistan stock market. *International Journal of Management Research & Emerging Sciences*, 10(2), 215-226.

Ashrafzadeh, M., Sadrani, M., & Zolfani, S. H. (2025). Clustering-based return prediction model for stock pre-selection in portfolio optimization. *Results in Engineering*, 27, Article 106263. <https://doi.org/10.1016/j.rineng.2025.106263>

Bali, T. G., Gokcan, S., & Liang, B. (2007). Value at risk and the cross-section of hedge fund returns. *Journal of Banking & Finance*, 31(4), 1135-1166.
<https://doi.org/10.1016/j.jbankfin.2006.10.023>

Bodnar, T., Lindholm, M., Niklasson, V., & Thorsen, E. (2022). Bayesian portfolio selection using VaR and CVaR. *Applied Mathematics and Computation*, 427, Article 127120.

Campbell, J. Y., & Viceira, L. M. (2002). *Strategic asset allocation: Portfolio choice for long-term investors*. Oxford University Press.

Chao, L. (2024). Application of machine learning in stock market return forecasting. *International Journal of Scientific Research and Management (IJSRM)*, 12(7), 6827-6835. <https://doi.org/10.18535/ijssrm/v12i07.em11>

Chaweewanchon, A., & Chaysiri, R. (2022). Markowitz mean-variance portfolio optimization with predictive stock selection using machine learning. *International Journal of Financial Studies*, 10(3), Article 64. <https://doi.org/10.3390/ijfs10030064>

Chen, R. (2025). Stock price prediction and portfolio optimization based on mean variance model and random forest model. *Advances in Economics, Business and Management Research*, 333, 358-367.

- Cheng, Y. (2025). Monte Carlo-Based VaR Estimation and Backtesting Under Basel III. *Risks*, 13(8), Article 146. <https://doi.org/10.3390/risks13080146>
- Chung, V., Espinoza, J., & Quispe, R. (2025). Forecasting Financial Volatility Under Structural Breaks: A Comparative Study of GARCH Models and Deep Learning Techniques. *Journal of Risk and Financial Management*, 18(9), Article 494. <https://doi.org/10.3390/jrfm18090494>
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253-263.
- Espiga-Fernandez, F., Garcia-Sanchez, A., & Ordieres-Mere, J. (2024). A Systematic Approach to Portfolio Optimization: A Comparative Study of Reinforcement Learning Agents, Market Signals, and Investment Horizons. *Algorithms*, 17(12), Article 570. <https://doi.org/10.3390/a17120570>
- Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2), 383-417. <https://doi.org/10.1111/j.1540-6261.1970.tb00518.x>
- Fan, Y. (2025). Enhancing investment strategies with LSTM-based stock prediction and mean-variance portfolio optimization. *Proceedings of the 3rd International Conference on Financial Technology and Business Analysis*. <https://doi.org/10.54254/2754-1169/2024.23667>
- Fapetu, O., Ojo, S. M., Balogun, A. A., & Asaolu, A. A. (2021). Capital market performance and macroeconomic dynamics in Nigeria. *FUOYE Journal of Finance and Contemporary Issues*, 1(1), 29-37.

- Fatouros, G., Makridis, G., Kotios, D., Soldatos, J., Filippakis, M., & Kyriazis, D. (2023). DeepVaR: A framework for portfolio risk assessment leveraging probabilistic deep neural networks. *Digital Finance*, 5(1), 29-56.
- Ferrari, D., Paterlini, S., Rigamonti, A., & Weissensteiner, A. (2024). Smoothed semicovariance estimation for portfolio selection. *Annals of Operations Research*. Advance online publication. <https://doi.org/10.1007/s10479-024-06043-z>
- Fleming, J., Kirby, C., & Ostdiek, B. (2001). *The economic value of volatility timing using 'realized' volatility* [Working paper]. Rice University, Jones Graduate School. <https://ssrn.com/abstract=276921>
- Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5), 2223-2273. <https://doi.org/10.1093/rfs/hhz113>
- He, W. (2022). An empirical research based on Markowitz's portfolio theory. *World Scientific Research Journal*, 8(3), 221-227. [https://doi.org/10.6911/WSRJ.202203_8\(3\).0028](https://doi.org/10.6911/WSRJ.202203_8(3).0028)
- Hsiao, Y.-Y. (2025). A comprehensive survey of modern portfolio optimization: From traditional risk analysis to advanced analytics and machine learning approaches (Preprint). *Johns Hopkins University*.
- Huang, R., Kambouroudis, D., & McMillan, D. G. (2025). Is portfolio diversification still effective: Evidence spanning three crises from the perspective of U.S. investors. *Journal of Asset Management*, 26(2), 115-135. <https://doi.org/10.1057/s41260-025-00398-z>
- Job, O. D. (2022). An empirical evaluation of alternative asset allocation policies for emerging and frontier market investors in Africa. *Journal of Financial Risk Management*, 11(3), 481-521. <https://doi.org/10.4236/jfrm.2022.113024>

- Jobson, J. D., & Korkie, B. M. (1981). Performance hypothesis testing with the Sharpe and Treynor measures. *The Journal of Finance*, 36(4), 889-908.
- Kevin, J., & Yugopuspito, P. (2025). Hybrid LSTM and PPO networks for dynamic portfolio optimization (LPPM-UPH, No. 404/LPPM-UPH/VII/2025). *Universitas Pelita Harapan Working Paper*.
- Leccadito, A., Staino, A., & Toscano, P. (2024). A novel robust method for estimating the covariance matrix of financial returns with applications to risk management. *Financial Innovation*, 10, Article 116.
- Ledoit, O., & Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850-859.
- Ledoit, O., & Wolf, M. (2011). Robust performance hypothesis testing with the variance. *Wilmott Magazine*, (55), 86-89.
- Li, G. (2023). Portfolio optimization and risk analysis in financial markets. *Proceedings of the 2nd International Conference on Financial Technology and Business Analysis*, 61, 243-247. <https://doi.org/10.54254/2754-1169/61/20231273>
- Lorimer, D. A., van Schalkwyk, C. H., & Szczygielski, J. J. (2024). Portfolio optimisation using alternative risk measures. *Finance Research Letters*, 67, Article 105758. <https://doi.org/10.1016/j.frl.2024.105758>
- Lyu, X. (2024). Portfolio Optimization Strategies: New Approaches Based on Machine Learning Forecasting. *Highlights in Business, Economics and Management*, 40, 1077-1082.

Mahadevaswamy, G. H., & Shyamala, G. (2022). Markowitz Model Is Right Choice to Investment. *International Journal of Novel Research and Development*, 7(11), 305-314.

Manogna, R. L., & Kulkarni, N. (2025). Portfolio Optimization Model for Stock Price Prediction Using Machine Learning. *Journal of Statistical Theory and Applications*, 24(1).

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77-91.
<https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>

Martinez-Barbero, X., Cervello-Royo, R., & Ribal, J. (2024). Portfolio optimization with prediction-based return using Long Short-Term Memory neural networks: Testing on upward and downward European markets. *Computational Economics*, 65, 1479-1504.
<https://doi.org/10.1007/s10614-024-10604-6>

Mba, J. C., Ababio, K. A., & Agyei, S. K. (2022). Markowitz mean-variance portfolio selection and optimization under a behavioral spectacle: New empirical evidence. *International Journal of Financial Studies*, 10(2), Article 28.
<https://doi.org/10.3390/ijfs10020028>

Memmel, C. (2003). Performance hypothesis testing with the Sharpe ratio. *Finance Letters*, 1(1), 21-23.

Michaud, R. O. (1989). The Markowitz optimization enigma: Is 'optimized' optimal? *Financial Analysts Journal*, 45(1), 31-42. <https://doi.org/10.2469/faj.v45.n1.31>

Moyoweshumba, E., & Seitshiro, M. (2025). Leveraging Markowitz, Random Forest, and XGBoost for optimal diversification of South African stock portfolios. *Data Science in Finance and Economics*, 5(2), 205-233. <https://doi.org/10.3934/DSFE.2025010>

- Mozumder, S., Hasan, M. K., & Kabir, M. H. (2024). An evaluation of the adequacy of Levy and extreme value tail risk estimates. *Financial Innovation*, 10(1), Article 100. <https://doi.org/10.1186/s40854-024-00614-6>
- Naeem, M., Jassim, H. S., & Korsah, D. (2024). The application of machine learning techniques to predict stock market crises in Africa. *Journal of Risk and Financial Management*, 17(12), Article 554. <https://doi.org/10.3390/jrfm17120554>
- Nahari, F. (2025). Portfolio optimization in practice: A comparative analysis of the Markowitz and Index models. In M. M. Husin (Ed.), *Proceedings of the 2025 International Conference on Financial Risk and Investment Management (ICFRIM 2025), Advances in Economics, Business and Management Research* (Vol. 333, pp. 367-375). Atlantis Press.
- Ojo, A. K., & Okafor, I. J. (2024). Forecasting Nigerian Equity Stock Returns Using Long Short-Term Memory Technique. *Journal of Advances in Mathematics and Computer Science*, 39(7), 45-54. <https://doi.org/10.9734/jamcs/2024/v39i71911>
- Okafor, C., & Robertson, A. (2022). Maximizing returns: Portfolio optimization in the Nigerian Stock Exchange. *International Journal of Advances in Applied Mathematics and Computer Science*, 10(2), 1-9.
- Quimbayo, C. Z., & Leon, B. (2025). Downside Risk Measures and ESG Factors in Optimal Portfolio Construction: Evidence from European Equity Markets. *Economics - Innovative and Economics Research Journal*, 13(2), 5-17. <https://doi.org/10.2478/eoik-2025-0082>
- Rigamonti, A., & Lucivjanska, K. (2021). Mean-semivariance portfolio optimization using minimum average partial. *SSRN Electronic Journal*, Article 3542727.

<https://doi.org/10.2139/ssrn.3542727>

Rockafellar, R. T., & Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3), 21-41. <https://doi.org/10.21314/JOR.2000.038>

Rockafellar, R. T., & Uryasev, S. (2002). Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7), 1443-1471. [https://doi.org/10.1016/S0378-4266\(02\)00271-6](https://doi.org/10.1016/S0378-4266(02)00271-6)

Sahiner, M. (2022). Forecasting volatility in Asian financial markets: Evidence from recursive and rolling window methods. *SN Business & Economics*, 2, Article 157. <https://doi.org/10.1007/s43546-022-00329-9>

Salo, A., Doumpos, M., Liesio, J., & Zopounidis, C. (2024). Fifty years of portfolio optimization. *European Journal of Operational Research*, 318(1), 1-18. <https://doi.org/10.1016/j.ejor.2023.12.031>

Samaniego Alcantar, A. (2023). Semi-variance optimization for the components of the Dow Jones Industrial Average index. *Contaduria y Administracion*, 68(4), 1-17. <http://dx.doi.org/10.22201/fca.24488410e.2023.3409>

San, J. (2025). Stock Forecasting and Portfolio Optimization Based on ARIMA-GARCH, Random Forest and Monte Carlo Models. *Advances in Economics, Business and Management Research*, 333, 580-588.

Senescall, M., & Low, R. K. Y. (2024). Quantitative Portfolio Management: Review and Outlook. *Mathematics*, 12(18), Article 2897. <https://doi.org/10.3390/math12182897>

Sulaiman, L. A., Adejayan, A. O., & Ilori, O. O. (2023). Capital Market Development and Economic Growth of West African Countries. *Nigerian Journal of Banking and*

Financial Issues, 9(1), 117-125.

Slusarczyk, D., & Slepaczuk, R. (2025). Algorithmic investment strategies on the Dow Jones Industrial Average. *Journal of Big Data*, 12, Article 127.

<https://doi.org/10.1186/s40537-025-01164-z>

Tjiwidjaja, H. (2025). Optimization of Investment Portfolio Returns Through an Integrated Risk Management Approach. *STIE Ganesha Research Papers*, 853-863.

Uzoaga, G. A., Adenomon, M. O., Nweze, N. O., & Maijama, B. (2025). Modelling and predicting stock prices of Nigerian Stock Exchange using some machine learning techniques and time series model. *Science World Journal*, 20(2), 510-515.

<https://dx.doi.org/10.4314/swj.v20i2.9>

Uzoaga, G. A., Adenomon, M. O., Nweze, N. O., & Maijamaa, B. (2025). Predictive machine learning methods for stock returns among emerging economies in Africa. *Science World Journal*, 20(3), 941-947. <https://dx.doi.org/10.4314/swj.v20i3.3>

Wahid, A. J., Riaman, & Sukono. (2025). A Systematic Literature Review on Mean-CVaR Based Financial Asset Portfolio Weight Allocation Using K-Means Clustering. *Journal of Applied Mathematics and Computing*, 10(2), 1069-1091.

Wang, T., Pan, Q., Wu, W., Gao, J., & Zhou, K. (2024). Dynamic Mean-Variance Portfolio Optimization with Value-at-Risk Constraint in Continuous Time. *Mathematics*, 12(14), Article 2268. <https://doi.org/10.3390/math12142268>

Yadav, A., Madhavi, R., Bagaria, O., Ambulkar, A., & Sharma, S. (2024). Survey on financial portfolio management's role in investment decision-making strategies.

Multidisciplinary Reviews, 6, Article e2023ss101.

<https://doi.org/10.31893/multirev.2023ss101>

- Yu, S. (2024). Advancing Stock Market Return Forecasting with LSTM Models and Financial Indicators. *Proceedings of the International Conference on Economic Management and Green Development*, Article 122. <https://doi.org/10.54254/2754-1169/122/2024.17734>
- Zaimovic, A., Arnaut-Berilo, A., & Bešlija, R. (2024). International Portfolio Diversification Benefits: An Empirical Investigation of the 28 European Stock Markets. *School of Economics and Business Sarajevo Working Papers*, Article 46.
- Zhang, G. (2025). Using Machine Learning for Stock Return Prediction. *Proceedings of ICEMGD 2025 Symposium*, Article LH23915. <https://doi.org/10.54254/2754-1169/2025.LH23915>
- Zhang, P., Yang, Y., Li, J., & Zeng, Y. (2023). A Two-stage Mean-CVaR Investment Strategy Based on LSTM. *Journal of South China Normal University (Natural Science Edition)*, 55(5), 93-102.
- Zhang, Y. (2024). Integrating Forecasting and Mean-Variance Portfolio Optimization: A Machine Learning Approach. *Proceedings of the 3rd International Conference on Financial Technology and Business Analysis*, Article 2002.
- Zsurkis, G., Nicolau, J., & Rodrigues, P. M. M. (2024). First passage times in portfolio optimization: A novel nonparametric approach. *European Journal of Operational Research*, 312(3), 1074-1085. <https://doi.org/10.1016/j.ejor.2023.07.044>